Supply Chain Management

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What is a Supply Chain?

It is the series of processes in the production and distribution of goods, from when they are first made, grown, etc. until they are bought or used. (Oxford dictionary)

Reverse supply chain: It is the series of activities required to retrieve a used product from a customer and either dispose of it or reuse it

> *"Competition is becoming less between individual companies, and more between the supply chains they are part of." -- Richard Wilding*

SUPPLY CHAIN

"The network of organizations that are involved, through upstream and downstream linkages, in the different processes and activities that produce value in the form of products and services in the hands of the ultimate customer." (Christopher 1998)

Christopher, M., 1999. Logistics and Supply Chain Management: Strategies for Reducing Cost and Improving Service Financial Times: Pitman Publishing. London, 1998

Process flow in a Supply Chain



Internal Supply Chain

The **internal supply chain** refers to all the supply chain activities that occur **within a single organization**, from the initial planning and procurement to production and internal distribution.

- **Purchasing**
- **Production:**
- **Sales**

•**Distribution:** Managing the flow of goods from the production facility to the end customer, including warehousing, transportation, and logistics.

•**Internal Integration:** Coordinating and integrating these functions to ensure a smooth and efficient flow of materials, information, and resources

Supply Chain Management (SCM)

- The term "Supply Chain Management" was first used in 1982 by Keith Oliver[1], a business consultant in an interview. In the 90s, this term gained popularity.
- SCM involves planning, management of activities in sourcing, procurement, conversion and logistic activities.

Logistics: The practical organization and detailed arrangements needed to make a complicated plan, involving a lot of people and equipment, successful. (Oxford dictionary)

[1]https://supplychaindigital.com/digital-supply-chain/keith-oliver-the-man-who-gave-us-supply-chain-management

Supply Chain Management (SCM)

- Proper planning, execution, control and monitoring of supply chain activities are done to ensure and create a net value, build a competitive environment, maintain a good balance between demand and supply.
- Proper coordination and collaboration is required with all the channels partners. Partners can be suppliers, intermediaries, third-party providers and end-user customers.

[1]https://supplychaindigital.com/digital-supply-chain/keith-oliver-the-man-who-gave-us-supply-chain-management

7R of Supply Chain Management

- **Right Product** – Delivering the correct product that meets customer needs.
- **Right Quantity** – Ensuring the exact amount of product is delivered, avoiding both shortages and excess.
- **Right Condition** – Making sure the product is delivered without damage and in usable condition.
- **Right Place** – Delivering the product to the correct location.
- **Right Time** – Ensuring timely delivery as per customer expectations.
- **Right Customer** – Making sure the product reaches the intended recipient.
- **Right Cost** – Providing the product at the most cost-effective rate, balancing quality and expense.

Objectives of Supply Chain Management

- Improving efficiency
- Improving quality
- Optimizing transportation and logistics
- Delivering in Competitive Pressures
- Managing inventories

Focus on (A) Lean (B) Agile (C) Leagile

Lean manufacturing is a philosophy and methodology focused on maximizing productivity and minimizing waste in manufacturing processes, aiming to create more value for customers with fewer resources.

Eliminating Waste (Muda):
Lean manufacturing identifies and eliminates activities that don't add value from the customer's perspective. This includes things like overproduction, waiting, excess inventory, unnecessary motion, defects, and underutilized talent

Mura means unevenness, non-uniformity, and irregularity. Mura is the reason for the existence of any of the seven wastes. Mura leads to Muda.

Muri means overburden, beyond one's power, excessiveness, impossible or unreasonableness. Muri can result from Mura and in some cases be caused by excessive removal of Muda (waste) from the process.

Agile Manufacturing

Agile manufacturing is a dynamic approach to production that prioritizes rapid responsiveness to changing customer needs and market demands, emphasizing speed and agility to gain a competitive edge.

Functions and Activities of SCM

- Forecasting
- Purchasing
- Inventory management
- Information management
- Quality assurance
- Scheduling
- Production and delivery
- Customer service

Benefits of Supply Chain Management

- Lower inventories
- Higher productivity
- Greater agility
- Shorter lead times
- Higher profits
- Greater customer loyalty
- Integrates separate organizations into a cohesive operating system

Evaluating Sources of Supply

- Vendor analysis: Evaluation is done in terms of
  - Price
  - Quality
  - Services
  - Location
  - Inventory policy
  - Flexibility

A tool of Supply Chain Management

- **Push/Pull strategy**
- **Pull system is the part of Just-in-time (JIT).**
- **Japanese term Kanban. English translation: Signboard.**
- Kanban: a Japanese manufacturing system in which the supply of components is regulated through the use of an instruction card sent along the production line (Oxford dictionary.)
- A kanban is a card of two types: (i) Withdrawal kanban (ii) Production Ordering kanban
- Withdrawal kanban shows the quantity that subsequent process should withdraw from the preceding one.
- Production ordering kanban shows the quantity that the preceding process should produce.

Bullwhip effect

The bullwhip effect is a supply chain phenomenon where small fluctuations in demand at the retail level cause progressively larger fluctuations at the wholesale, distributor, and manufacturer levels, leading to inefficiencies and increased costs.

Causes of Bullwhip Effect

- **Lack of communication and disorganization**
  - This may happen between each supply chain link when managers reduce product demand differently within different links of the supply chain and order different quantities.
  - It leads to disorganization and hinders the smooth working of supply chain processes.

- **Order batching**
  - Companies sometimes don't place orders with the supplier after receiving an order.
  - This alters the variability in demand. Sometimes there could be a surge while other times there could be a dip in demand.

Causes of Bullwhip Effect

- **Variation in pricing**
  - Promotional discounts, special offers can disrupt the usual demand for products. It leads to inaccurate demand forecasting and consequently over-production.

Bullwhip Effect can be minimized by

Better communication between managers, improve inventory planning process, Collaboration between customers and suppliers

Ethics in Supply Chain Management (SCM)

An ethical supply chain is a practice that focuses on the need for corporate social responsibility, working to produce products and services in a way that treats its workers and the environment ethically. Supporting an ethical supply chain means that companies will incorporate social and human rights and environmental considerations into how they do business across the world. (https://www.the-future-of-commerce.com/2020/01/22/ethical-supply-chain-definition-stats/)

Material Requirements Planning (MRP) & Manufacturing Resource Planning (MRP-II)

An Overview of Key Concepts and Applications

Introduction to MRP

- What is MRP?
- - A system for calculating material requirements based on production schedules.
- - Developed in the 1960s to address inventory management challenges.

- Key Goals:
- 1. Ensure the right materials are available at the right time.
- 2. Minimize inventory costs.
- 3. Optimize production schedules.

Components of MRP

- 1. Master Production Schedule (MPS):
- - Specifies what to produce and when.
- 2. Bill of Materials (BOM):
- - A structured list of materials and components required for production.
- 3. Inventory Status File:
- - Tracks inventory levels and records.

Working of MRP

- Input:
- - Master Production Schedule (MPS)
- - Bill of Materials (BOM)
- - Inventory Status File

- Process:
- - Explodes BOM, calculates net material requirements, and schedules production.

- Output:
- - Material plans, purchase orders, and work orders.

Introduction to MRP-II

- What is MRP-II?
- - An extension of MRP that integrates other business processes like financials, capacity planning, and human resources.
- - Introduced in the 1980s to create a more comprehensive resource planning system.

- Key Goals:
- - Optimize manufacturing resources (materials, manpower, machines).
- - Enhance decision-making through data integration.

Key Features of MRP-II

- 1. Capacity Planning:
- - Matches resource availability with production requirements.
- 2. Financial Integration:
- - Links production planning with cost management.
- 3. Sales and Operations Planning (S&OP):
- - Aligns demand forecasts with production schedules.
- 4. Feedback Loops:
- - Adjusts plans based on actual performance and variances.

Comparison of MRP and MRP-II

- Feature | MRP | MRP-II
- ------- | --- | ------
- Focus | Material planning | Comprehensive resource planning
- Scope | Limited to production needs | Includes financials, HR, etc.
- Integration | Standalone system | Integrated with ERP systems
- Feedback | No real-time feedback | Real-time adjustments possible

Benefits of MRP and MRP-II

- MRP Benefits:
- - Reduces inventory levels.
- - Improves order accuracy.

- MRP-II Benefits:
- - Enhances resource utilization.
- - Facilitates data-driven decision-making.
- - Improves cross-functional collaboration.

Challenges of MRP and MRP-II

- MRP Challenges:
- - Dependency on accurate data.
- - Cannot handle capacity constraints.

- MRP-II Challenges:
- - High implementation cost.
- - Requires extensive employee training.
- - Complexity of integration.

Applications of MRP and MRP-II

- Industries:
- - Automotive, aerospace, electronics, and manufacturing sectors.

- Software Solutions:
- - Examples include SAP, Oracle, and Microsoft Dynamics.

What is a Digital Supply Chain?

- In a **typical (traditional) supply chain**, the flow of goods involves
  - Designing a product, sourcing and procuring the raw materials and parts
  - making the product, estimating demand, planning how the firm will market the final product, taking orders, arranging logistics and sales channels and
  - Finally providing customers with visibility into their orders.

Items tend to travel linearly, with each step dependent on the previous one. While conceptually simple, the linear pre-digital approach to supply chains is dependent on each step in the process performing about as expected.

What is a Digital Supply Chain?

- A **digital supply chain**, in contrast, to a traditional supply chain provides more visibility into the workings of the chain. The added near real-time visibility of supplier performance and customer needs lets supply chain owners develop more complex relationships with more suppliers.
- Digital supply chains are more customer-centric and aim to meet today's three pillars of excellence in demand fulfillment: **speed, personalization and choice**.

What is a Digital Supply Chain?

- The technology is typically used to monitor manufacturing or logistics processes. Through the use of gateways, that data can be shared with stakeholders along the supply chain.
- The data is then aggregated and used by stakeholders to assess whether they can reliably expect the products they need.

https://www.netsuite.com/portal/resource/articles/erp/digital-supply-chain.shtml#target2

Internet of Things

- Internet of Things (IoT) is defined as a network of different physical objects that are embedded with digital devices such as sensors and actuators.
- The elements are connected in order to collect and exchange data.
- Furthermore, communication is enabled between objects or ‘things’, systems, and services in the network that allows for data sharing.

❑ The term “Things” refers to any entity that are embedded with sensors and software with the purpose of connecting and exchanging data with other systems.

Source: Govindan, K., Kannan, D., Jørgensen, T.B. and Nielsen, T.S., 2022. Supply chain 4.0 performance measurement: A systematic literature review, framework development, and empirical evidence. Transportation Research Part E: Logistics and Transportation Review, 164, p.102725.

Internet of Things

- IoT is the result of many technological developments complementing each other and allowing for the gap between the physical and virtual worlds to be bridged.
- The capabilities from the different developments include sensing, actuation, communication, cooperation, addressability, identification, embedded information processing, localization, and user interfaces.
- The network infrastructure created by the IoT that connects the virtual systems and physical objects allows for more efficient, accurate, and intelligent operations performance

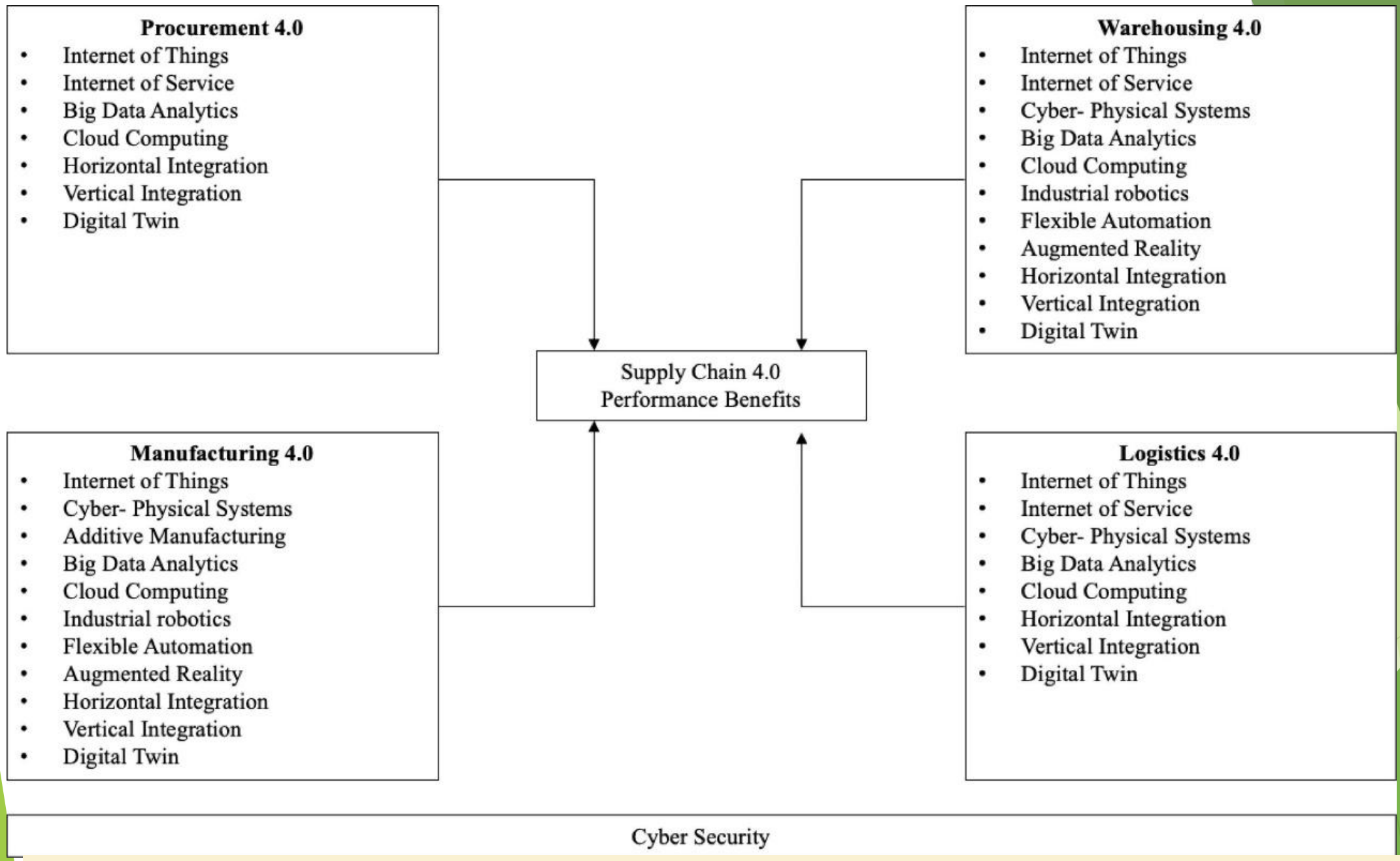
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IoT in Smart Logistics

- IoT is considered the critical enabler for smart logistics services.
- Smart logistics management of objects and information flows cannot be facilitated without the IoT tracking and connecting the status of moving objects in the supply chain.
- Moving objects include goods, trucks and storage capacity.
- Immediate collection and efficient monitoring of data associated with the physical objects enables smart logistics.

IoT in Smart Logistics

- As the amount of bar codes, sensors, and RFID tags equipped to physical objects increases, the logistics and transportation companies' ability to monitor the movement of these objects in real time throughout the supply chain increases.
- Transportation management systems and IoT will be a core part of the transportation and logistics industries in the future. IoT will heavily affect automobile services and transportation systems.
- **EXAMPLE**: Improved sharing of under-utilized resources among vehicles through enhanced sensing, data processing, networking, and communication capabilities by tracking and monitoring the location of each vehicle and its movement and predicting the future location of the vehicle.



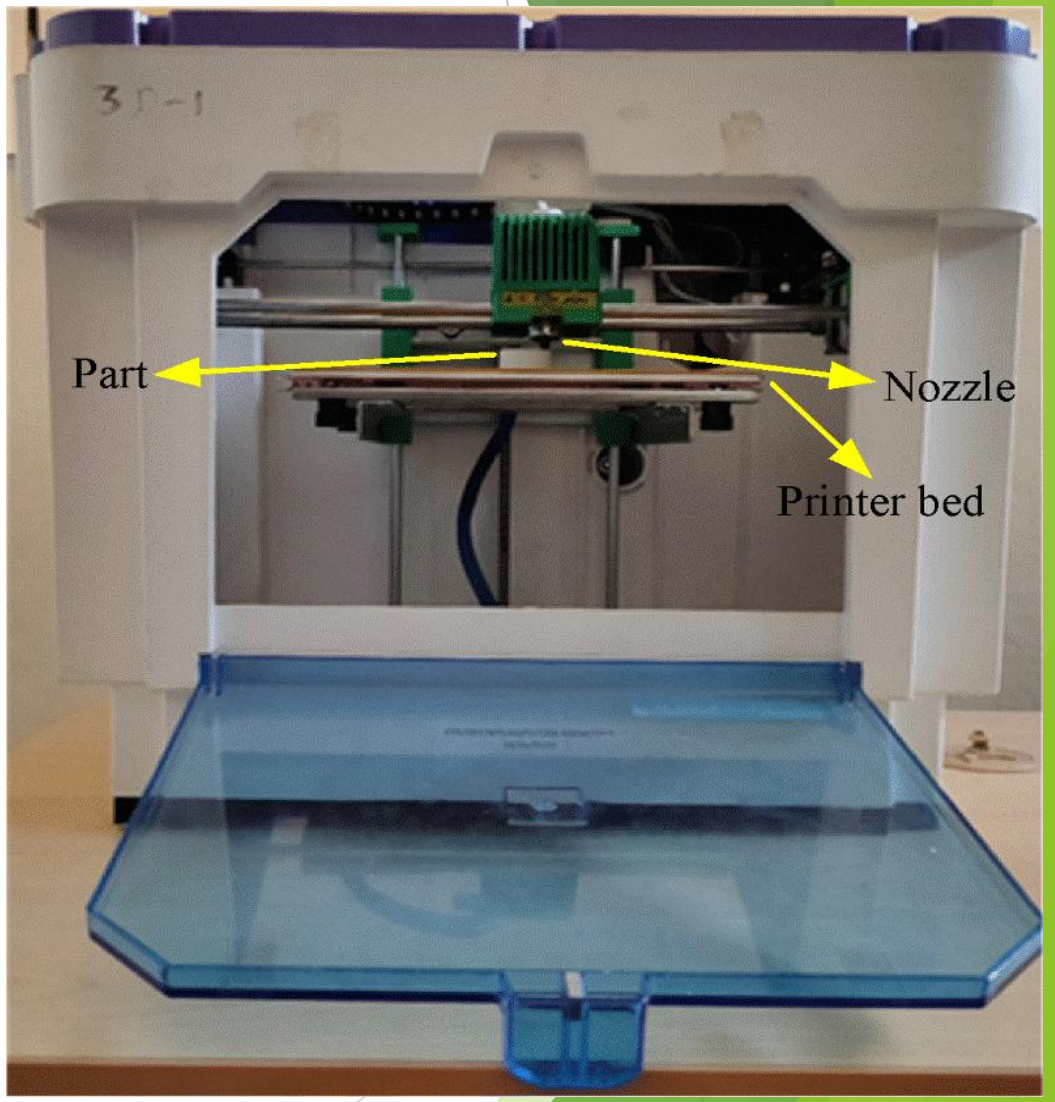
Source: Govindan, K., Kannan, D., Jørgensen, T.B. and Nielsen, T.S., 2022. Supply chain 4.0 performance measurement: A systematic literature review, framework development, and empirical evidence. Transportation Research Part E: Logistics and Transportation Review, 164, p.102725. *(under a creative commons)*

How to develop a digital supply chain?

- **Define Digital Strategy & Vision**
  Align supply chain digitization with business goals; identify key technologies (AI, IoT, cloud, etc.).
- **Ensure Data Integration & Connectivity**
  Connect systems across suppliers, operations, and customers; enable real-time data flow.
- **Implement Automation & Smart Operations**
  Use AI, RPA, and robotics for forecasting, planning, and efficient operations.
- **Enable Real-Time Visibility**
  Monitor activities through dashboards and digital twins; detect and respond to disruptions quickly.
- **Foster Partner Collaboration**
  Integrate suppliers and logistics partners digitally; enhance data sharing and coordination.
- **Drive Continuous Improvement with Analytics**
  Use data insights and AI to optimize performance and adapt to changes dynamically.

Impact of 3D printing on Supply Chain

- 3D printing is a part of digital manufacturing and Industry 4.0.
- Based on a digital CAD model, 3D Printing technology creates a physical object by adding layers of material in succession.
- It lowers the cost by cutting out waste, unlike subtractive manufacturing.
- It reduces carbon footprint and improves service by producing customized products.

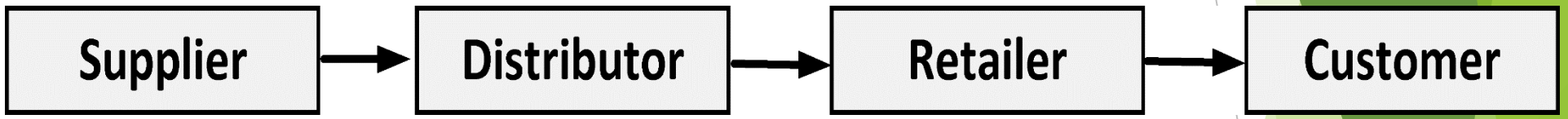


Fused Deposition Modelling based 3D Printing

Impact of 3D Printing on the supply chain



Conventional Supply Chain



3D Printing Distributed Supply Chain (Distributors procure 3D printers)



3D Printing Retailer Supply Chain (Retailers procure 3D printers)